

2019

COMPUTER SCIENCE — PRACTICAL

Paper : CSM - 105P

(Module - I)

[Data Structure]

Full Marks : 50

Marks Distribution :

Sessional : 15

Attendance : 05

Experiment : 25

Viva voce : 05

Instruction

1. The examinee must write the problem statement in answer script.
2. The examinee must write the algorithm clearly in pseudo language with maximum number of comments.
3. The examinee must write all the assumptions, if any.
4. The examinee must write the justification of the data structure.
5. The examinee must write the results along with the inputs in a suitable format.
6. The examinee must write the discussions along with the pros and cons of the implementation.
7. The examinee should take the printout of the code and it should be attached with the answer script. If possible, the examinee should take the screen shot of input and output of the given problem.
8. The answer script must be signed by the examiner.

Answer *one question* only.

1. Write and implement an algorithm to differentiate the edges of a directed graph following the notion of depth-first search, based on your sequence of visiting the vertices. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
2. Write and implement an algorithm to compute the number of components of an undirected graph. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
3. Write and implement an algorithm to differentiate the edges of an undirected graph following the notion of depth-first search, based on your sequence of visiting the vertices. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
4. Write and implement an algorithm that evaluates a given expression in its infix form. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
5. Write and implement an algorithm that determines whether a stream of parentheses is valid in normal algebraic sense. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
6. If the elements in the band formed by the three diagonals of a tridiagonal matrix X are represented column-wise in an array Y , with $X[1, 1]$ being stored in $Y[1]$, then write and implement an algorithm to determine the value of $X[i, j]$, $1 \leq i, j \leq n$ from array Y .

7. Write and implement an algorithm to compute the shortest distance and the path between a pair of vertices of an undirected graph, based on your sequence of visiting the vertices. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
8. Write and implement an algorithm to compute the shortest distance and the path between a pair of vertices of a directed graph, based on your sequence of visiting the vertices. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
9. Let X be an $n \times n$ matrix, where $3 \leq n \leq 12$, such that the elements in each row are in non-decreasing order but the elements in each column are in non-increasing order. Write and implement a decision tree structure based algorithm that finds the first three largest elements in X and their sequences in non-increasing order in a minimum number of comparisons.
10. Devise a scheme in computing a polynomial C , where C is computed by (i) multiplying two polynomials A and B , and (ii) differentiating polynomial A . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.
11. Devise a scheme in representing a sparse matrix A and transpose this representation of A in lexicographic order. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
12. Write and implement an algorithm to construct an equivalent binary tree of the following forest that comprises T_1 and T_2 . Clearly show the necessary steps of the construction.
 - (a) $T_1 : (A(B(E,F),C(G),D(H(K),I(L, M),J))))$, and
 - (b) $T_2 : (N(O(P)))$.

13. Devise a scheme in computing a polynomial C , where C is computed by (i) adding two polynomials A and B , and (ii) subtracting polynomial B from polynomial A . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.
14. Let X and Y be two lower triangular matrices, each with n rows. Devise and implement a scheme to represent both the triangles in an array Z of size $n \times (n + 1)$. Write algorithms to determine the values of $X[i, j]$ and $Y[i, j]$, $1 \leq i, j \leq n$ from array Z . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.
15. Write and implement an algorithm that computes a B-tree of order m and perform insertion of an element into the tree as well as deletion of an element from the tree. As for example, consider the keys to be inserted are : $a_5, g_2, f_1, b_3, k_3, d_6, g_4, m_5, j_2, e_3, s_1, j_1, r_5, z_7, b_6, m_2, m_9, t_3, u_5, q_6$ into an initially empty tree, in the order given, when the value of $m = 5$. Also delete the keys one after another, from the final B-tree obtained, in the following order : g_4, r_5, q_6, d_6 and obtain subsequent B-trees.
16. Let A be an $n \times n$ matrix that contains non-zero elements present only in five diagonals, centring and including the major diagonal. If the elements in the band formed by these five diagonals are represented column wise in an array B , with $A[1, 1]$ being stored in $B[1]$, obtain an algorithm to determine the value of $A[i, j]$, $1 \leq i, j \leq n$ from array B . Choose appropriate data structure(s) for the said purpose, and make clear why and how the data structure(s) is (are) suitable.
17. Write and implement an algorithm that determines the location of a saddle point, if one exists, in a matrix of size $m \times n$. Devise a scheme to make your implementation more efficient. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
18. Consider a sequence S of $n \geq 15$ elements and implement *ShellSort* of S . Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

19. Consider a binary search tree and take necessary steps to make it an AVL tree, if needed. Perform deletion of elements from the AVL tree and make the tree AVL after deletion of each element from the tree. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

 20. Consider the problem of implementing n -digit binary integers and compute amortized cost, a_i , of the problem in terms of n . Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

 21. Consider a sequence of n elements and construct an AVL tree after insertion of each element into the tree. Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

 22. Consider a sequence S of $n \geq 15$ elements and implement *HeapSort* of S . Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.

 23. Consider a sequence S of $n \geq 15$ elements and implement *SieveSort* of S . Choose appropriate data structure(s) for the said purpose and make clear why and how the data structure(s) is (are) suitable.
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